

Whether Token Probabilities Catch a Wrong Tool Call Depends on the Model and the Kind of Error

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Abstract

Teams that deploy tool-calling language models want to stop a wrong call before it executes, using the token probabilities every serving stack returns or a probe on the model’s hidden states. Earlier comparisons of the two held out items rather than tools and pooled every kind of error, so they cannot say which errors the cheaper signal misses. We compare them on held-out tools, on six open-weight checkpoints of up to three billion parameters from five families and on three benchmarks, separately for each kind of error, beside a length floor and two judges that read only the output. On wrong argument values the probe beats the mean token log-probability on Llama-3.2 and Gemma-3 by up to +0.357 in area under the curve, and confidence beats the probe on every Qwen3, Qwen3.5 and MiniCPM5 run, even when these models are forced to write JSON. On dropped parallel calls the probe leads confidence on every run with enough of them. Much of the probe’s lead is visible in the text of the call, since a second small model reading the call comes within noise of the probe on most runs where the probe leads. Before building an internal judge, an operator should score the model’s confidence per kind of error and train a judge on the output text first.

1. Introduction

A language model agent acts by calling tools. When it calls the wrong tool, or the right one with wrong arguments, the call is usually well formed, so the runtime executes it before anyone reads the transcript. A guardrail therefore has to judge the call between its generation and its execution, from what the model exposes in that forward pass. Two signals are available there. The output distribution gives the

probability of each generated token, which every serving stack returns. The hidden states give a probe something to read, which only a stack that runs the model’s own code can expose.

Probes on the residual stream separate wrong calls from right ones with high accuracy (Healy et al., 2026; Yeats et al., 2026), and a few neurons suffice to detect invalid arguments, unneeded calls and missing calls (Ke et al., 2026). An earlier comparison by Anonymous (2026) put token probabilities beside these probes on Glaive and found the probabilities near chance and head-averaged attention spectra strong. That comparison split the data by prompt rather than by tool and pooled every kind of error. The question this paper answers is which wrong tool calls a model’s own token probabilities miss, on which models, and whether a judge trained on the output text alone would catch what they miss.

We answer it with one quantity, the internal advantage: the area under the curve (AUC) of a probe on the residual stream minus that of the mean token log-probability of the call, on tools never seen in training, beside a floor that reads only lengths. We compute it on every failure and on two kinds of error, and we set the probe against two trained comparators that need no internals: an output-only judge fitted on the same labels and a second language model that reads the request and the call.

The result fits in one sentence: whether token probabilities catch a wrong tool call depends on the model and on the kind of error (Figure 1). On wrong argument values the probe leads confidence by +0.134 to +0.357 on six of the seven Llama-3.2 and Gemma-3 runs, and confidence leads the probe on all six Qwen3, Qwen3.5 and MiniCPM5 runs, native or forced into JSON. We call this the value-error split. On dropped parallel calls the probe leads confidence on all four runs that have at least twenty of them. Forcing JSON makes Qwen3.5 and MiniCPM5 drop such calls, which is why our registered prediction about forced JSON failed. Much of what the probe reads is in the output text: an output-only judge recovers most of the probe’s lead on four of the seven probe-side runs, and a Qwen3.5-0.8B reader of the call is within noise of the probe on six of the seven probe-side runs where it was fitted.

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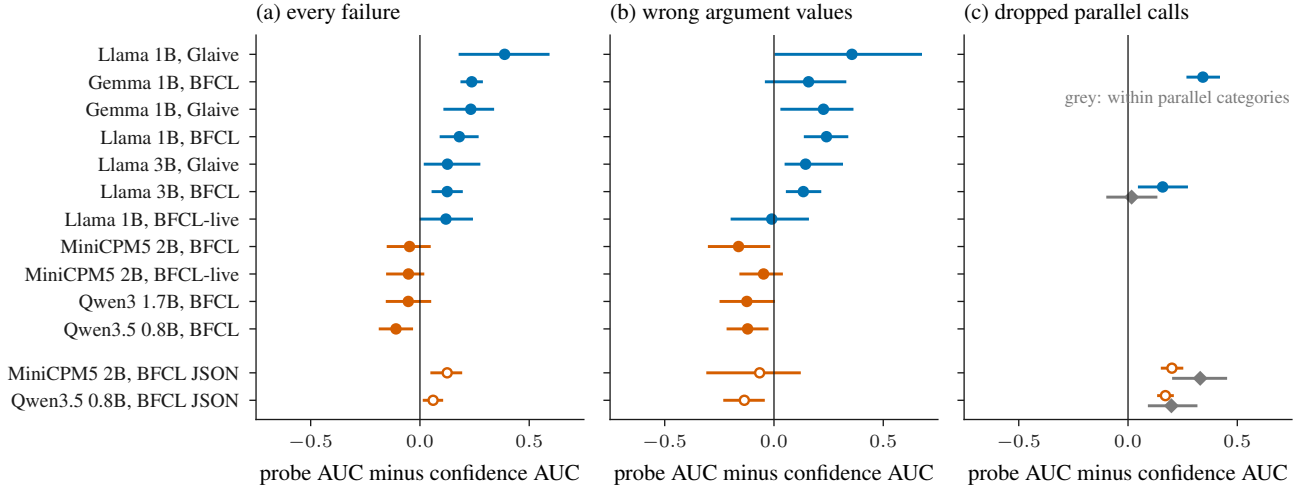


Figure 1. Token-role probe AUC minus mean log-probability AUC on held-out tools, per run, dot at the mean over five split seeds, bar the 95% interval from resampling tools. (a) Every failure against valid calls. (b) Wrong argument values against valid calls. (c) Dropped parallel calls against valid calls, coloured, and within the parallel categories only, grey. Blue Llama-3.2 and Gemma-3, orange Qwen3, Qwen3.5 and MiniCPM5, open markers the forced-JSON runs. A run appears in (b) and (c) when it has at least twenty failures of that kind.

What follows in practice is an order of operations (Section 8): score confidence per kind of error on held-out tools first, train a judge on the output text before reading internals where confidence misses value errors, and check dropped parallel calls separately.

Contributions.

1. The value-error split, measured on 11 powered runs from six checkpoints in five families on held-out tools, with a length floor and a paired interval on every difference (Section 4). Healy et al. (2026), Yeats et al. (2026) and Ke et al. (2026) establish that internal detectors work. This paper adds the decomposition by kind of error against the model’s own confidence, on unseen tools.
2. A correction of Anonymous (2026): on held-out tools and repaired labels confidence reaches 0.576 to 0.823 AUC, head-averaged attention readouts sit near the length floor, and size does not separate the sides (Sections 4 and 7).
3. The comparators that bound the claim: an output-only judge and a reader model under the same protocol, which show where the probe’s lead is readable from the text (Section 5).
4. On the theory side, a causal two-head construction showing that no readout of the head-averaged attention graph can separate head configurations whose per-head connectivity differs by a length-independent margin, and that under the normalised Laplacian the gap between the two changes sign with length (Section 7, Section A).

2. Related work

Probes on hidden states. Supervised probes on the residual stream recover whether a statement is true (Azaria & Mitchell, 2023; Marks & Tegmark, 2024), predict hallucination risk from the query alone (Ji et al., 2024) and stand in for sampling-based uncertainty at the cost of one pass (Kossen et al., 2024). Orgad et al. (2025) locate the signal at the exact-answer tokens, which is the observation the token-role probe of Healy et al. (2026) applies to a call’s name, arguments and closing delimiter. Yeats et al. (2026) train linear probes on eighteen models on BFCL and find that model size predicts probe accuracy first and probing depth second. Ke et al. (2026) find small sets of neurons that detect invalid arguments, unneeded calls and missing calls on six models. These results establish that internal detectors of wrong calls work. Healy et al. (2026) and Yeats et al. (2026) hold out items, so a tool seen in training can return at test, and neither reports the model’s own token probabilities on the same items.

Token probabilities against probes on tool calls. Anonymous (2026) compared the mean token log-probability with the token-role probe and with head-averaged spectral probes on Glaive under splits by prompt, and asked whether the best detector changes with model scale. They found token probabilities near chance and head-averaged spectra strong. Under the tool-held-out splits and the repaired labels of Section 3, neither finding holds (Sections 4 and 7): confidence ranges from 0.576 to 0.823 AUC, head-averaged readouts sit near the length floor, and the side a model falls on does not change from Llama-3.2-1B to 3B. This paper adds the decomposition by kind of error, the output-only comparators

and three further families.

Readouts of attention. LapEigvals trains a probe on the top eigenvalues of the attention Laplacian (Binkowski et al., 2025), Lookback Lens on the share of attention paid to the context (Chuang et al., 2024), LLM-Check on eigenvalues of attention kernels (Sriramanan et al., 2024) and SinkProbe on attention sinks (Binkowski et al., 2026). Noël (2025) introduced graph-signal diagnostics on the Laplacian of a head-averaged attention graph, and Dahlem et al. (2026) prove that every transpose-invariant spectral diagnostic is blind to the direction of attention flow. We reimplement LapEigvals, Lookback Lens and SinkProbe after their authors’ papers and code, and score them under the probe’s classifier, floor and held-out tools. No numerical parity check against the original implementations is reported, so their numbers here are those of our reimplementations.

Output confidence. Token probabilities are informative about a model’s own correctness on question answering (Kadavath et al., 2022), and sampling-based estimates sharpen them (Farquhar et al., 2024; Manakul et al., 2023) at a cost in forward passes that an agent about to act does not have. Tool-calling benchmarks measure whether a call is right (Patil et al., 2025; 2024; Qin et al., 2024) and not whether the model’s confidence tracks it. Our contribution is to put the single-pass confidence of the model beside single-pass judges on its internals and on its output, per kind of error, on unseen tools.

3. Setup

Task, labels and kinds of error. A model receives a user request and a set of tool schemas through its own chat template and generates greedily. The call is parsed in whichever format the model writes, JSON for Llama-3.2, Gemma-3 and Qwen3 and an XML dialect for Qwen3.5 and MiniCPM5, and compared with the benchmark’s ground truth after both are normalised to a name and an argument dictionary. Each call receives one failure mode: valid, wrong name, wrong argument values, missing arguments, missing parallel calls, an unneeded call, no call, an unparseable call, or a call cut off by the budget. Every number is computed on the scored population, the items where a call is required and the generation can be judged. We report each detector on every failure and on two kinds of error against the valid calls alone: wrong argument values, where the right tool gets a wrong value, and dropped parallel calls, where a request needs several calls and the model writes fewer.

Conditioning facts. Two facts shape every later number. First, the labelling and extraction code was audited and repaired before the measurements here (Section C). The

largest repair concerns correct list-valued arguments, which had been scored wrong on the JSON formats: on Qwen3-1.7B it turned 158 labelled failures into 52. Relabelling the stored generations of all 16 runs with the released labeller reproduces every stored label. Second, some failures are schema echoes, calls whose arguments copy the tool’s JSON schema. A pattern for a copied `properties` field or object type finds them in 73% of Gemma-3 failures on BFCL, in 34% and 38% on two Llama-3.2-1B runs, and on no Qwen3, Qwen3.5 or MiniCPM5 run. Most are labelled missing arguments and stay out of the value-error comparison. The pattern misses partial echoes, so these shares are lower bounds (Section H).

Models and benchmarks. The checkpoints are Llama-3.2-1B and 3B (released September 2024), Gemma-3-1B (March 2025), Qwen3-1.7B (April 2025), Qwen3.5-0.8B (February 2026) and MiniCPM5-2B (September 2026), with dense, sliding-window and hybrid linear-attention stacks (Section L). The chat templates of the last three carry a reasoning mode, which is disabled at generation through-out. The benchmarks are the Glaive function-calling corpus (GlaiveAI, 2024), the single-turn categories of BFCL-v4 (Patil et al., 2025) and its live categories, contributed by users. A run is one checkpoint on one benchmark. A run with fewer than thirty items of either class is underpowered and excluded from every aggregate, and a kind of error is scored on a run only when it has at least twenty failures of that kind.

Detectors. Confidence is the mean token log-probability of the generated span, with its direction fixed on training items, chosen before any result was seen. The probe is the token-role probe of Healy et al. (2026), a regularised logistic regression on the residual state at the function name, the argument values and the closing delimiter, at eight depths. Two trained comparators need no internals of the deployed model. The output-only judge is one logistic regression on what an operator sees from outside the model: lengths, benchmark category, counts read off the call, every stored confidence summary, and character n-grams of the call and the request. The reader is Qwen3.5-0.8B, which reads the request and the call as plain text, without schemas, and is probed with the same classifier. The floor is a classifier on prompt length, generation length and truncation.

Protocol and quantity. Items are grouped by ground-truth tool and split five ways at the group level, so every score is on tools absent from training. Every trained detector is fitted on the training folds restricted to the scored population, with a validation carve-out of training tools. Each item is scored once out of fold, one pooled AUC is computed per split seed, and five seeds are averaged. Every difference between detectors is a paired bootstrap that resamples whole tools,

with its 95% percentile interval. The internal advantage of a run is the AUC of the probe minus that of confidence. Computed on value errors against valid calls, it defines the two sides of the split: the probe side, where it is positive, and the confidence side, where it is negative.

4. The internal advantage by kind of error

Table 1 and Figure 1 hold the paper’s measurement. On every failure, the seven powered Llama-3.2 and Gemma-3 runs have an internal advantage of +0.119 to +0.387, with an interval excluding zero on six of them. The four powered Qwen3, Qwen3.5 and MiniCPM5 runs read −0.110 to −0.048, with an interval below zero on one. Runs that share a checkpoint are not independent, so the unit of any test is the checkpoint. With three against three checkpoints the smallest attainable two-sided rank-test value is $p = 0.100$, so the separation is a description of these six checkpoints.

Wrong argument values. The pooled comparison mixes kinds of error whose sign differs, so Figure 1b scores each against valid calls alone. On wrong argument values the probe leads confidence on six of the seven probe-side runs, by +0.134 to +0.357, with an interval excluding zero on five. The exception is Llama-3.2-1B on the live categories at −0.010. Confidence leads the probe on all six confidence-side runs, native and forced into JSON, where the advantage is −0.162 to −0.048. On these models the mean log-probability already ranks most wrong values below right ones. Over every failure, an equal-weight fusion of probe and confidence gains −0.003 to +0.034 over confidence alone on the confidence side, with no interval excluding zero, against +0.108 to +0.290 on the probe side.

Dropped parallel calls. On all four runs with at least twenty dropped calls the probe leads confidence, by +0.158 to +0.342 (Figure 1c). Valid calls outside the parallel categories cannot drop a call, so part of that lead is a category contrast. Within the parallel categories the lead holds on MiniCPM5-2B and Qwen3.5-0.8B under forced JSON, +0.330 [+0.20, +0.45] and +0.198 [+0.09, +0.32], and vanishes on Llama-3.2-3B at +0.016 [−0.10, +0.13]. Gemma-3 makes 5 valid parallel calls and cannot be tested there.

Output format. Qwen3 writes JSON like Llama-3.2 and Gemma-3 and sits on the confidence side, so the native format does not decide the side. Registered prediction R3 said that, forced to write JSON, the confidence-side families keep a pooled advantage at or below zero, and that it fails above +0.10. It failed. MiniCPM5-2B moved to +0.125 [+0.05, +0.19] and Qwen3.5-0.8B to +0.060 [+0.01, +0.11]. The shift is carried by dropped parallel calls, which make up 56 of 83 failures and 143 of 226 under forced JSON. On

wrong argument values both models stay on the confidence side (Table 1, last rows). The forced format changed which errors the models make and left their confidence on value errors where it was.

Size and release date. The probe side holds checkpoints of one and three billion parameters and the confidence side checkpoints of 0.8 to two. Within Llama-3.2 the probe does not improve from 1B to 3B, 0.831 at both sizes on BFCL and 0.963 against 0.911 on Glaive, and the side does not change. Two properties of the model set do separate the sides on these six checkpoints: release date, with a cutoff between March and April 2025, and a reasoning mode in the chat template. Section 8 returns to both.

Calls with a history and replication. With earlier ground-truth calls in the prompt, on the multi-turn categories of BFCL-v4, MiniCPM5-2B reads 0.900 for the probe and 0.859 for confidence, an advantage of +0.041 [−0.01, +0.09] opposite in sign to its single-turn runs, with tools and conversations linked into four groups. Its single-turn side is not reproduced with a history (Section K). Llama-3.2-1B on BFCL, extracted twice before and after the extraction repairs, reads +0.224 and +0.180. Differences smaller than this drift of 0.044 are not interpreted.

5. What the probe adds over the output text

The internal advantage sets a trained probe against an untrained summary of the output, so part of the probe’s lead could come from training on labels rather than from the hidden states. The comparators of this section are trained on the same labels, folds and training population as the probe and read only what is visible outside the deployed model.

The output-only judge. On the confidence side the output-only judge scores 0.558 to 0.678, below confidence itself, because a few dozen failures cannot fit its n-gram features. On the probe side it is strong (Figure 2a). It recovers 57 to 95% of the probe’s lead over confidence on four of the seven runs, all of them Llama-3.2-1B or Gemma-3 runs. Probe minus output-only judge is +0.011 to +0.227 and excludes zero on four runs, against six for probe minus confidence. On Gemma-3 on BFCL the judge reaches 0.900 against the probe’s 0.948, and a structural floor that counts schema keywords already reaches 0.916. On Llama-3.2-3B the judge trails the probe by +0.227 on Glaive and +0.199 on BFCL.

A second model reading the call. A reader that is a different language model, given the request and the generated call as plain text, is a stronger output-side judge than n-grams. On the seven probe-side runs, probe minus reader ranges from −0.020 to +0.074, and its interval contains zero on

Table 1. Every run under the corrected labels. n_+ and n_- are the labelled failures and valid calls in the scored population. The AUC columns are the length floor, the mean log-probability (conf.), the output-only judge, the Qwen3.5-0.8B reader and the token-role probe. The last four columns are differences of AUC on held-out tools: probe minus confidence with its 95% tool-resampled interval, probe minus the output-only judge, and probe minus confidence on wrong argument values and on dropped parallel calls, each against valid calls. ° the interval excludes zero (intervals in Figures 1 and 2 and Sections F and G). n.c. not computed: fewer than twenty failures of that kind, an underpowered run, or a comparator not fitted on that run. * re-extracted under the corrected extractor. † fewer than thirty items of one class.

Model	Data	n_+	n_-	floor	conf.	judge	reader	probe	probe – conf.	probe – judge	value err.	dropped
Llama-3.2-1B	Glaive	221	359	0.539	0.576	0.910	0.949	0.963	+0.387 [+0.18, +0.59]	+0.053	+0.357°	n.c.
Llama-3.2-1B*	BFCL	161	339	0.635	0.651	0.755	0.818	0.831	+0.180 [+0.09, +0.27]	+0.077°	+0.240°	n.c.
Llama-3.2-1B	BFCL-live	208	157	0.709	0.663	0.682	0.795	0.782	+0.119 [−0.00, +0.24]	+0.101°	−0.010	n.c.
Llama-3.2-3B	Glaive	90	555	0.522	0.785	0.684	0.931	0.911	+0.126 [+0.02, +0.28]	+0.227	+0.145°	n.c.
Llama-3.2-3B	BFCL	120	576	0.618	0.707	0.633	0.758	0.831	+0.125 [+0.05, +0.20]	+0.199°	+0.134°	+0.158°
Gemma-3-1B	Glaive	158	460	0.704	0.650	0.872	0.853	0.883	+0.233 [+0.11, +0.34]	+0.011	+0.226°	n.c.
Gemma-3-1B*	BFCL	364	150	0.842	0.711	0.900	0.927	0.948	+0.237 [+0.19, +0.29]	+0.048°	+0.158	+0.342°
Qwen3-1.7B	BFCL	52	639	0.628	0.799	0.558	0.662	0.745	−0.053 [−0.16, +0.05]	+0.187°	−0.124	n.c.
Qwen3-1.7B†*	Glaive	10	699	0.677	0.867	n.c.	n.c.	0.676	−0.191 [−0.65, +0.21]	n.c.	n.c.	n.c.
MiniCPM5-2B*	BFCL	54	640	0.578	0.823	0.585	0.677	0.775	−0.048 [−0.15, +0.05]	+0.190°	−0.162°	n.c.
MiniCPM5-2B*	BFCL-live	120	475	0.545	0.797	0.642	0.655	0.744	−0.053 [−0.15, +0.02]	+0.102°	−0.048	n.c.
MiniCPM5-2B†*	Glaive	9	716	0.685	0.913	n.c.	n.c.	0.710	−0.203 [−0.51, +0.01]	n.c.	n.c.	n.c.
Qwen3.5-0.8B*	BFCL	86	398	0.601	0.792	0.678	0.730	0.682	−0.110 [−0.19, −0.03]	+0.004	−0.121°	n.c.
MiniCPM5-2B*	BFCL, JSON	83	514	n.c.	0.757	n.c.	n.c.	0.882	+0.125 [+0.05, +0.19]	n.c.	−0.066	+0.201°
Qwen3.5-0.8B*	BFCL, JSON	226	436	n.c.	0.808	n.c.	n.c.	0.868	+0.060 [+0.01, +0.11]	n.c.	−0.136°	+0.171°

six of them (Figure 2b). On all three Llama-3.2-1B runs and on Llama-3.2-3B on Glaive the reader is within two points of the model’s own probe (Table 1). The one run where the probe beats every output-side judge is Llama-3.2-3B on BFCL, by +0.074 [+0.01, +0.14] over the reader. On the other probe-side runs the probe shows that the error is visible in the call. The claim that the model’s states hold what its confidence lacks, beyond what its text shows, rests on that one run. On the confidence side the reader scores 0.655 to 0.730, below the model’s own confidence with an interval excluding zero on three of the four runs.

What the probe adds. Fusing the probe with the output-only judge by an equal-weight rank average, with no fitted weight, gains +0.021 to +0.161 over the judge alone on the probe side, with an interval excluding zero on five of seven runs. The value-error split itself does not depend on any of these comparators, because it compares the probe with the model’s own confidence on one kind of error.

6. Controls on the split

A difference between model families can come from the items, the labels or the protocol. Each paragraph below changes one thing and keeps the rest of the protocol fixed.

Item difficulty. A model that fails rarely fails on hard items, and its confidence may register hardness instead of its own error. On BFCL an item’s difficulty is the share of the other checkpoints that fail it. That share alone predicts a model’s failures at 0.697 to 0.782 AUC. Scoring both de-

tectors inside strata of difficulty removes what either gains from knowing which items are hard, and the split does not close (Figure 3a). The advantage is +0.182 on Llama-3.2-1B and +0.182 on Llama-3.2-3B, and between −0.099 and −0.062 on the three confidence-side checkpoints. Confidence on the confidence side tracks the model’s own errors beyond the hardness of the request.

Label budget. A probe needs labelled failures and confidence needs none, and the confidence-side models fail less often. Training the Llama-3.2 probes on as few failures as the other families supply, with the test folds untouched, costs them 0.041 AUC at 1B and 0.018 at 3B (Figure 3b). At 51 training failures the advantage is still +0.137 and +0.110, above every value on the other side.

The confidence summary. The mean log-probability is one summary of the output distribution. On the runs that store up to 11 length-invariant summaries, the summary with the highest AUC on the other folds is applied fold by fold, so no summary is chosen on the items it scores (Figure 3c, Section I). The advantage becomes +0.165 on Llama-3.2-1B and +0.330 on Gemma-3 on BFCL, and at most −0.032 on the confidence side. An oracle that picks the best summary on the test items leaves the probe side at +0.119 or more.

Benchmark and schema echoes. Gemma-3 reads +0.233 on Glaive and +0.237 on BFCL, Llama-3.2-1B keeps its side on three benchmarks and MiniCPM5 on two, so no run’s side depends on its benchmark. With matched

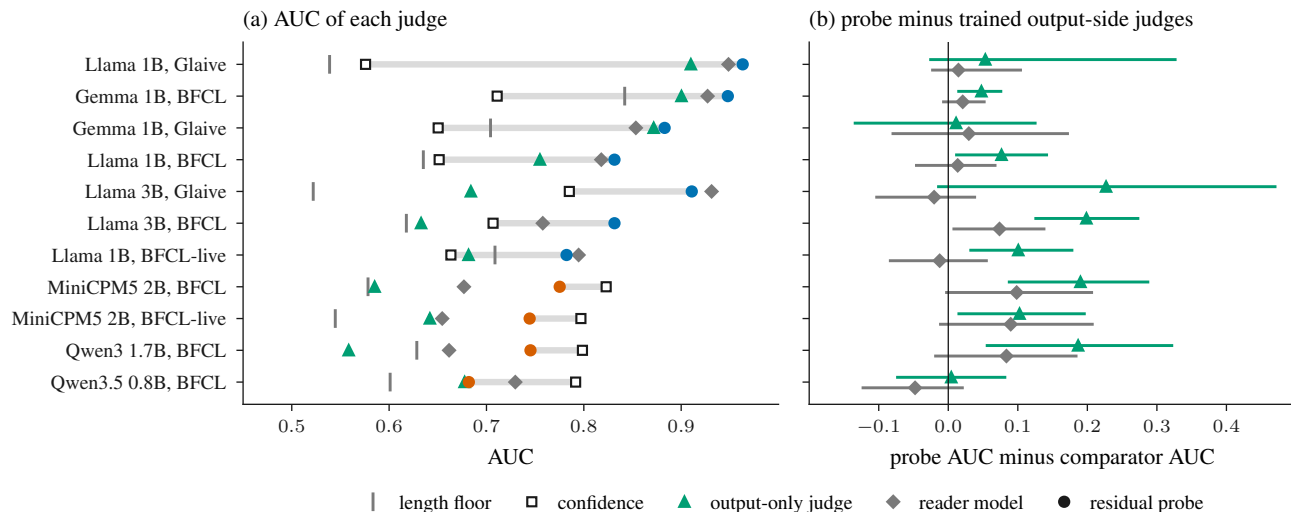


Figure 2. (a) AUC per powered run of the length floor, the mean log-probability, the output-only judge, the Qwen3.5-0.8B reader and the token-role probe, with the span from confidence to probe shaded. (b) Probe AUC minus output-only judge AUC (triangles) and minus reader AUC (diamonds), with 95% tool-resampled intervals. Reader points appear on the runs where the reader was fitted.

schema echoes removed from evaluation, the pooled advantage keeps its sign on six of the seven probe-side runs. The exception is Llama-3.2-1B on the live categories, which moves from $+0.119$ to -0.020 $[-0.16, +0.10]$. Its value-error advantage is also near zero, so this run is on neither side once format failures are set aside.

Chance level of the pipeline. With the labels permuted, 20 refits of the whole probe pipeline on Qwen3.5-0.8B score 0.509 AUC with a standard deviation of 0.042. The standard deviation of the permuted advantage is 0.057, which is why single-run advantages of a few hundredths are not interpreted. Inside individual folds the advantage is negative on zero to three of 25 fold and seed pairs on the probe side and on 16 to 24 on the confidence side.

Training population and pooling. Training the probe on the scored population only, and fixing the sign of the log-probability on that population, both moved the numbers toward confidence (Section C). Pooling out-of-fold scores across folds penalises a fitted probe and not a fixed score: the advantage computed within folds exceeds the pooled one by $+0.018$ on average.

7. Attention readouts, head averaging and cost

A stack may expose attention weights but not activations. On the probe-side runs, the per-head spectral profile and LapEigvals trail the token-role probe by a few points (Figure 4a), for instance $+0.066$ on Llama-3.2-1B and $+0.066$ on Gemma-3 on BFCL. The two attention readouts are within noise of each other, and on Llama-3.2-1B and Gemma-3 on BFCL, where it was computed, the anchored

readout of the call’s own rows ties the per-head profile ($+0.007$ and $+0.037$).

Head averaging. Whether an attention readout works depends on whether heads are averaged before anything is computed. The head-averaged spectral profile, the usual read of a layer in this literature and the one Anonymous (2026) found strong, sits within -0.065 to $+0.117$ of the length floor on the probe-side runs. On the stored per-head features of 11 runs, keeping heads apart scores 0.790 against 0.709 for the same metrics averaged over heads, a gain that is positive on 11 of 11 runs. Permuting the head axis independently per item, which keeps every value and destroys only which head produced it, costs $+0.195$, and padding the averaged features with noise to the per-head width does not recover it (Section J).

The construction behind this is elementary, and Section A states it for causal attention. Take a previous-token head and a sink head, both causal and row-stochastic. Replacing both by their average gives a second configuration with the same head-averaged graph. Any statistic of the averaged graph returns the same value on both, while the spread of per-head algebraic connectivity is zero in one and tends to one half in the other as the sequence grows (Proposition 1). For the combinatorial Laplacian the averaged graph can only overstate the mean connectivity over heads (Proposition 2). The normalised Laplacian the detectors use has no such bound: in the same construction at $T = 200$ the averaged graph reads 0.334 against a per-head mean of 0.500, and on 210 measured layers the bound fails on three.

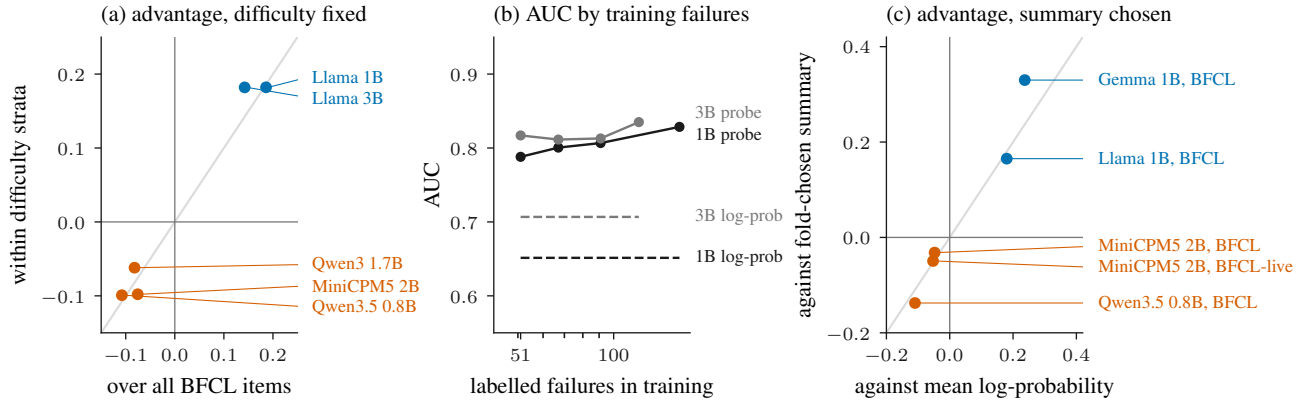


Figure 3. (a) Internal advantage within strata of item difficulty against the advantage over all BFCL items, one point per checkpoint, identity line grey. Difficulty is the share of the other checkpoints that fail the item. (b) Token-role probe AUC (solid) and mean log-probability AUC (dashed) against the labelled failures available for training on Llama-3.2-1B and 3B, log scale. (c) Internal advantage against a confidence summary chosen on training folds and against the mean log-probability, one point per run that stores several summaries, identity line grey.

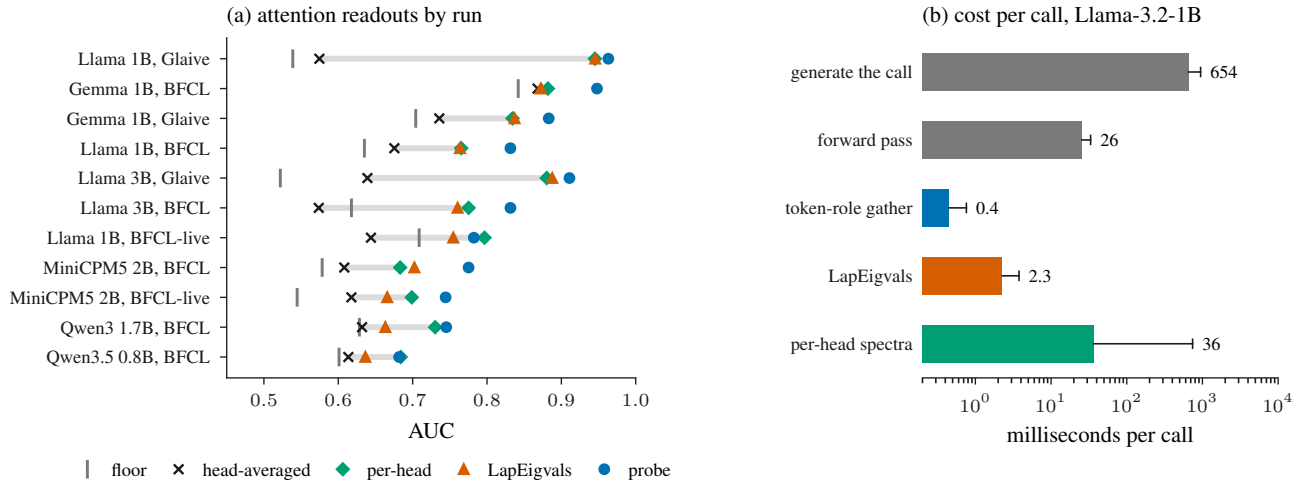


Figure 4. (a) AUC per powered run of the length floor, the head-averaged spectral profile, the per-head spectral profile, LapEigvals and the token-role probe, with the span from head-averaged to per-head shaded. (b) Cost per call of each path on Llama-3.2-1B, bar at the median over 20 BFCL prompts, line to the 90th percentile, log scale, one consumer GPU.

Cost. Reading confidence costs nothing beyond generation. Every internal judge needs a teacher-forced pass over prompt and call, 26 ms against 654 ms of generation (Figure 4b). On top of the pass the token-role features cost 0.4 ms and LapEigvals 2 ms. The per-head spectra cost 36 ms at the median and 741 ms at the 90th percentile, more than generation when the call is long. The output-only judge needs no pass of the deployed model, and the reader model needs one pass of Qwen3.5-0.8B over the request and the call.

8. Discussion

Readings of the split. Two properties of the model set fit all six checkpoints (Table 3). One is release date, with a

cutoff between March and April 2025. The other, registered as hypothesis H1 before any new model was extracted (Section D), is the post-training recipe: the three confidence-side families ship a chat template with a reasoning mode and were post-trained with it, and the two probe-side families do not. We read the split as possibly a post-training effect: training that rewards a model for working through a call before writing it may also leave its token probabilities tracking its own value errors. The reading is untested here. Switching the reasoning mode on at generation within Qwen3-1.7B yields 21 failures in 396 scored calls, too few to score, and Qwen3 is already on the confidence side with the mode off. A third reading, that a model which has seen the benchmark in training fails less and knows when it fails, is weakened by the live categories, contributed by users after the curated

set, on which MiniCPM5 stays on the confidence side.

What confidence misses. On the probe side the mean log-probability ranks wrong values close to chance on BFCL, 0.587 on Llama-3.2-1B and 0.552 on Gemma-3, while the residual stream separates them. We interpret this as an output distribution that does not express what the model’s states, and on most runs its own text, carry. Section 5 bounds the reading.

For operators. The decision that changes is which judge to build, if any, in order of cost.

Before building an internal judge for a tool-calling model

1. Score the mean token log-probability per kind of error on a few hundred labelled calls from the model to be deployed, on tools absent from the labelled set. Where it already ranks value errors (Table 1, confidence side), a probe fused with it gained at most +0.034 AUC over every failure.
2. Where it misses value errors, train an output-only judge or a small reader model on the same labels before reading internals. The output-only judge recovered 57 to 95% of the probe’s lead on four of seven probe-side runs and needs no access to the model’s states.
3. Check dropped parallel calls separately, by counting the calls a parallel request needs. On every run with at least twenty of them the probe led confidence by +0.158 or more.

For model builders. If a release reported the internal advantage per kind of error on held-out tools, an operator could see before deployment whether the model’s confidence flags its value errors and its dropped calls.

For detector research. Under splits by prompt and pooled labels, Anonymous (2026) found token probabilities near chance and head-averaged spectra strong. Under splits by tool, with the labels repaired and the errors separated, confidence reaches 0.576 to 0.823 AUC and beats the probe on value errors in three of the five families. A probe’s accuracy reported without the model’s own confidence, without an output-only judge and without held-out tools can show a strong detector on a model that did not need one.

9. Conclusion

Whether a tool-calling model’s token probabilities catch its wrong calls depends on the model and on the kind of error. On wrong argument values the probe leads confidence on Llama-3.2 and Gemma-3 and trails it on Qwen3, Qwen3.5 and MiniCPM5, whatever format the latter are made to write. On dropped parallel calls the probe leads wherever there are enough of them. Much of the probe’s lead is visible in the text of the call to a judge that never reads the model’s internals. Three recommendations follow, in order of cost:

score the model’s confidence per kind of error on held-out tools before building any judge, train a judge on the output text before a probe where confidence misses value errors, and check dropped parallel calls separately.

10. Limitations

The first limitation is the instrument. Labels come from comparing a call with a benchmark’s ground truth, which is a proxy for whether the action would have been harmful, and Section C shows that a parser can manufacture a family effect. The hand audit after the second repair found 18 errors in 240 items, all correct calls scored as failures, and the residual after the third repair is unmeasured. Errors of that kind favour probes, which can learn the surface form of a mislabelled call, so residual label error would move runs toward the probe side.

The second is the comparator. The protocol repairs of Section 6 moved numbers toward confidence, and comparing a trained probe with an untrained confidence score moves them toward the probe: against the output-only judge the probe’s lead falls to +0.011 to +0.227, with an interval excluding zero on four of seven probe-side runs. The reader sees no tool schemas, so it understates what the text reveals. The output-only judge is weak on the confidence-side runs, where few failures and many n-gram features let it overfit, so it does not bound confidence there.

The third is scale and breadth. Every checkpoint has at most three billion parameters and the two sides hold three checkpoints each, so the split is a description of these checkpoints and no checkpoint-level test can separate it. Release date and a reasoning mode in post-training fit the division equally well, and Yeats et al. (2026) find probe accuracy rising with size, so the side a family falls on above this range is open.

The fourth is realism. Calls are single-step on three benchmarks of English requests. The multi-turn replay on one model did not reproduce that model’s single-turn side and rests on four linked groups. The forced-JSON MiniCPM5 run covers part of BFCL and was extracted from a tree with uncommitted changes. Cost is measured on one consumer GPU, and the attention baselines are our reimplementations.

Impact statement

This work measures open-weight language models on public tool-calling benchmarks, trains no language model and involves no human subjects. Its aim is defensive: to tell an operator which wrong tool calls a model’s own confidence already flags and which ones need another judge before the call executes. The same detectors could guide the construction of calls that evade them, which is why we report every

kind of error and every run on which a detector weakens rather than one headline number.

Reproducibility statement

Every number in this paper expands from a macro that a released script builds from the result files, and a provenance file records the file, field and run each macro reads. Labels are recomputed from the stored generations at evaluation time, so the labelling rule in force is the one in the released code, and relabelling the stored generations of 16 runs under it reproduces every stored label. Splits are grouped by ground-truth tool and every interval resamples tools. Section D dates each registered prediction against the labelling repairs that produced the data testing it, and R1 was written when the split under the unrepaired labels was already known. Models and benchmarks are public, and the stored extractions the tables are computed from are released with the code.

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A. Head averaging under causal attention

Purpose: the two statements behind Section 7, proved for the causal attention of the decoders studied here, and the scope of each.

Fix one layer with H heads and T positions. Head h produces a row-stochastic attention matrix A_h , lower triangular under a causal mask. It is symmetrised and stripped of self-loops into $W_h = \frac{1}{2}(A_h + A_h^\top) - \text{diag}(\frac{1}{2}(A_h + A_h^\top))$, with combinatorial Laplacian $L_h = D_h - W_h$ and symmetric normalised Laplacian $\mathcal{L}_h = I - D_h^{-1/2}W_hD_h^{-1/2}$. The head-averaged graph is $\bar{W} = \frac{1}{H} \sum_h W_h$, and λ_2 is the second-smallest eigenvalue.

Proposition 1 (head averaging is many-to-one). *Any map of the form $\{W_h\} \mapsto \Phi(\bar{W})$ is constant on head configurations sharing the same mean. For every $T \geq 3$ there are two configurations of two causal row-stochastic heads with the same mean whose spreads of per-head connectivity, $\max_h \lambda_2(L_h) - \min_h \lambda_2(L_h)$, differ by $\cos(\pi/T) - \frac{1}{2}$.*

Proof. The first claim holds because Φ factors through $\bar{W}$. For the second, let A_P be the previous-token head, $A_P[i, i-1] = 1$ for $i \geq 1$ and $A_P[0, 0] = 1$, and A_S the sink head, $A_S[i, 0] = 1$ for every i . Both are causal and row-stochastic, and so is $A_M = \frac{1}{2}(A_P + A_S)$. Symmetrisation and the removal of the diagonal are linear, so $W_M = \frac{1}{2}(W_P + W_S)$, and the configurations $\{A_P, A_S\}$ and $\{A_M, A_M\}$ share the mean graph W_M . W_P is the path on T nodes with edge weights $\frac{1}{2}$, whose Laplacian has eigenvalues $1 - \cos(k\pi/T)$, so $\lambda_2(L_P) = 1 - \cos(\pi/T)$. W_S is the star centred on position 0 with edge weights $\frac{1}{2}$, whose Laplacian has eigenvalues 0, $\frac{1}{2}$ with multiplicity $T - 2$, and $T/2$, so $\lambda_2(L_S) = \frac{1}{2}$ for $T \geq 3$. The spread is $\cos(\pi/T) - \frac{1}{2}$ in the first configuration and zero in the second. $\square$

Proposition 2 (head averaging overstates combinatorial connectivity). *For any heads $W_1, \dots, W_H$, $\lambda_2(\bar{L}) \geq \frac{1}{H} \sum_h \lambda_2(L_h)$, with equality if and only if the heads admit a common Fiedler vector.*

Proof. The Laplacian is linear in the adjacency, so $\bar{L} = \frac{1}{H} \sum_h L_h$. By the min-max theorem of Courant and Fischer, $\lambda_2(L) = \min_{x \perp \mathbf{1}, \|x\|=1} x^\top L x$ is a pointwise minimum of linear functionals of L and hence concave. If x^* attains the minimum for $\bar{L}$ then $\lambda_2(\bar{L}) = \frac{1}{H} \sum_h x^{*\top} L_h x^* \geq \frac{1}{H} \sum_h \lambda_2(L_h)$, with equality exactly when x^* is a Fiedler vector of every head (Chung, 1997). $\square$

Scope. Both statements hold for any row-stochastic heads, so the causal decoders here are instances of them, and Proposition 1 uses only causal heads. Proposition 2 covers the combinatorial Laplacian only. The spectral detectors of Section 7 use the normalised Laplacian, whose Rayleigh quotient is not linear in W , and for it no inequality in either direction holds. In the construction of Proposition 1, computed by `analysis/v2-causal-construction.py`, the normalised connectivity of the averaged graph minus the per-head mean is +0.090 at $T = 5$ and -0.166 at $T = 200$, so the sign of the gap changes with length. Under the combinatorial Laplacian the same difference is +0.116 at $T = 5$ and +0.0001 at $T = 200$, as Proposition 2 requires. On 210 measured layers from three models and three inputs the combinatorial inequality holds without violation, the normalised one fails on three layers, and the averaged graph reports connectivity a median 14% above the mean over heads. Neither statement predicts which models fall on which side of the value-error split.

Remark 1 (LapEigvals reads degrees). Binkowski et al. (2025) observe that for a causal attention matrix the Laplacian $L = D - A$ is lower triangular, so its spectrum is its diagonal $\{d_j - A_{jj}\}$. Their top- k features depend on A only through its

column sums and diagonal. The symmetrisation above destroys triangularity, so the per-head profile reads global functionals of the routing. The two families are not two estimates of one object, and what they share is that both keep heads apart.

B. Protocol details

Purpose: every setting needed to rerun the extraction and the scoring without another paper.

Generation. Greedy decoding in bfloat16 with deterministic kernels, a budget of 256 new tokens (160 on multi-turn prompts), tool schemas passed through the model’s chat template with its reasoning mode disabled, and every end-of-turn id of the template among the stopping ids. The rendered text is tokenised without a second beginning-of-sequence token. Each record stores its budget, library versions and commit.

Features. Residual states are read at eight depths spaced evenly over the layer count at the function-name token, the mean over argument-value tokens and the closing delimiter, located by aligning the parsed call to token offsets in each dialect. Attention features are reduced inside a forward hook on each attention module, on the subgraph induced on the generated call.

Output-only judge and reader. The structural features are the floor’s three lengths, a one-hot benchmark category, and counts read off the generated call: calls, arguments, characters, digits, quoted strings and schema keywords. The confidence features are every stored summary of the token log-probabilities. The text features are character 2 to 5-gram TF-IDF vectors of the call and of the request. The output-only judge concatenates all three. The reader model Qwen3.5-0.8B reads the request, a newline and the generated call, without the tool schemas, and its residual state at eight depths is read at the last token and averaged over the call’s tokens. Both are fitted with the probe’s classifier on the probe’s folds, training population and validation carve-out, and the folds are asserted equal to the stored ones. As a reverse check the same code recomputes the paper’s length floor, which matches the stored floor on every run to within $4e-3$.

Classifiers and scoring. Logistic regression with ℓ_2 penalty, standardised features, class balancing and $C \in \{0.01, 0.1, 1, 10\}$ chosen on a one-fifth validation carve-out of training tools. Five tool-grouped folds, training restricted to the scored population, one pooled AUC per seed over five seeds, mean reported. Differences are paired bootstraps that resample whole tools, 2000 draws per seed for the stored contrasts and 1000 for the audit contrasts, draws pooled over seeds, percentile intervals.

C. Labelling and extraction defects found by the audit

Purpose: the repairs that condition every number, with the side each defect favoured.

Defect	Where it fell	Favoured
A predicted list argument was collapsed to its first element by the unwrapping meant for ground truth, so correct list-valued calls were scored wrong	JSON formats (Llama-3.2, Qwen3). 158 → 52 failures on Qwen3-1.7B	probes, through a surface feature
Semicolon-separated parallel calls did not parse and were scored unparseable	Llama-3.2-3B, 185 calls returned to the scored population	probes
A list sent as a string was compared inconsistently with a list	Llama-3.2	probes
Probe positions never located in the MiniCPM5 dialect, so the probe read the end-of-sequence state for every role	MiniCPM5	confidence
Generation stopped only on end-of-text, so Qwen3.5 ran past its end of turn	Qwen3.5	unclear, small
A second beginning-of-sequence token	Llama-3.2, MiniCPM5	neither
Probes trained on every item and scored on a subset learned format failures the subset excludes	population including abstentions	probes
The sign of the log-probability fixed on items it is never scored on	Llama-3.2, two runs	probes
Runs sharing a checkpoint treated as independent in a rank test	aggregate	the apparent significance

Table 2 reports a hand-checked sample of thirty items per run drawn on the labels after the first two repairs, judged against the benchmark’s function documents. Every error found was a correct call scored as a failure, 18 of 240 sampled items, from two causes the benchmark’s own checker handles: list arguments written as single-quoted Python literals, and mathematical expressions that differ only in notation. Both are repaired in the released labeller, which applies the checker’s string normalisation to both sides, and every number in this paper is computed under that third repair. The residual error after the third repair has not been sampled by hand. Errors of the kind found inflate probes, which can learn the surface form of a mislabelled correct call.

Table 2. Hand audit of the labels after the first two repairs, thirty items per run stratified over failure modes, Wilson intervals. All errors are correct calls scored as failures.

Model	Data	n	errors	rate [95% interval]	dominant cause
Llama-3.2-1B	BFCL	30	5	0.17 [0.07, 0.34]	python literal not decoded
Llama-3.2-3B	BFCL	30	6	0.20 [0.10, 0.37]	python literal not decoded
Qwen3-1.7B	BFCL	30	2	0.07 [0.02, 0.21]	notation caret
Gemma-3-1B	BFCL	30	0	0.00 [0.00, 0.11]	none
MiniCPM5-2B	BFCL	30	4	0.13 [0.05, 0.30]	notation caret
Qwen3.5-0.8B	BFCL	30	1	0.03 [0.01, 0.17]	notation caret
Llama-3.2-1B	Glaive	30	0	0.00 [0.00, 0.11]	none
Gemma-3-1B	Glaive	30	0	0.00 [0.00, 0.11]	none

D. Registered predictions and outcomes

Purpose: every prediction committed to the repository, beside its outcome. C1 was committed before the audit, R1 to R3 before the first labelling repair, and H1 and H2 after the first two repairs and before the third. R1 was written when the split under the unrepaired labels was already known. P1 and P2 were written on the day of the October audit, before the measurement.

Id	Prediction	Outcome
R1	The internal advantage separates the families by release period: MiniCPM5 and Qwen3.5 below Llama-3.2, Gemma-3 and Qwen3 at every checkpoint.	Failed. Qwen3-1.7B moved to the confidence side once the list-argument defect was repaired. The division that holds was found afterwards and is descriptive.
R1b	The tool-tuned Llama-xLAM-2-8b has a smaller internal advantage than Llama-3.1-8B-Instruct, with an interval excluding zero.	Not run.
R2	On a distractor-tool dose ladder the internal advantage rises with the number of distractors on at least one model from each side.	Not run.
R3	Under forced JSON output the families where confidence wins keep an advantage at or below zero, and the prediction fails if it exceeds +0.10.	Failed on MiniCPM5-2B, +0.125 [+0.05, +0.19], on a partial run of 616 of 850 items extracted from a tree with uncommitted changes. Undecided on Qwen3.5-0.8B, +0.060 [+0.01, +0.11]. The shift is carried by dropped parallel calls. On wrong argument values both stay below zero (Section 4).
C1	The per-head spectral score and LapEigvals are not interchangeable, each keeps signal when the other is conditioned on, and their union exceeds both.	Two parts held and the third failed (union gain within noise on eleven runs), measured on the labels before the audit.
H1	Models whose post-training included a reasoning mode have an internal advantage lower by at least 0.05 than a matched model without it, same base.	No matched pair has been run. Qwen3-1.7B with its reasoning mode switched on at generation fails on 21 of 396 scored calls, below the thirty needed to score.
H2	Within one model the advantage stays within 0.05 of its full value when its labelled failures are cut to what the other families supply and when distractor tools are added.	The label half holds (Figure 3b). The distractor half is not run.
P1	On dropped parallel calls, the entropy of the step at which the model stops beats the mean log-probability on every run with enough of them.	Failed. It held on Gemma-3, +0.155, and on Qwen3.5-0.8B under forced JSON, +0.057, and went the other way on MiniCPM5-2B under forced JSON, −0.214.
P2	On wrong argument values the stop-step entropy adds nothing over the mean log-probability.	Held on seven of seven runs.

E. Every detector on every run

Purpose: the AUC of every detector, so that any comparison in the body can be recomputed. SinkProbe and the anchored readout were computed only on runs extracted after these detectors were added, and the other cells read n.c.

Model	Data	token-role	token-level	head-avg.	per-head	LapEigvals	Lookback	SinkProbe	anchored	log-prob	floor
Llama-3.2-1B	Glaive	0.963	0.972	0.575	0.945	0.945	0.943	n.c.	n.c.	0.576	0.539
Llama-3.2-1B	BFCL	0.831	0.798	0.675	0.765	0.764	0.758	0.789	0.773	0.651	0.635
Llama-3.2-1B	BFCL-live	0.782	0.765	0.644	0.797	0.755	0.808	n.c.	n.c.	0.663	0.709
Llama-3.2-3B	Glaive	0.911	0.861	0.639	0.880	0.888	0.892	n.c.	n.c.	0.785	0.522
Llama-3.2-3B	BFCL	0.831	0.786	0.574	0.775	0.760	0.753	n.c.	n.c.	0.707	0.618
Gemma-3-1B	Glaive	0.883	0.898	0.736	0.834	0.837	0.786	n.c.	n.c.	0.650	0.704
Gemma-3-1B	BFCL	0.948	0.934	0.868	0.882	0.872	0.923	0.919	0.919	0.711	0.842
Qwen3-1.7B	BFCL	0.745	0.655	0.632	0.730	0.663	0.711	n.c.	n.c.	0.799	0.628
Qwen3-1.7B [†]	Glaive	0.676	0.790	0.445	0.781	0.678	0.709	0.702	0.674	0.867	0.677
MiniCPM5-2B	BFCL	0.775	0.775	0.608	0.683	0.702	0.744	0.754	0.764	0.823	0.578
MiniCPM5-2B	BFCL-live	0.744	0.742	0.618	0.699	0.666	0.732	0.706	0.700	0.797	0.545
MiniCPM5-2B [†]	Glaive	0.710	0.760	0.635	0.713	0.652	0.723	0.690	0.773	0.913	0.685
Qwen3.5-0.8B	BFCL	0.682	0.717	0.614	0.684	0.636	0.680	0.708	0.756	0.792	0.601

F. Output-only judges and the reader model per run

Purpose: each output-side comparator of Section 5 on each run, with its paired interval against the probe. Structural floor, confidence summaries in one regression and text n-grams are the three parts of the output-only judge. The fusion is an equal-weight rank average of the probe and the output-only judge, with no fitted weight. The reader cells read n.c. where the reader run is absent.

Model	Data	struct.	conf. LR	text	judge	probe — judge	fusion — judge	reader	probe — reader
Llama-3.2-1B	Glaive	0.557	0.493	0.916	0.910	+0.053 [−0.03, +0.33]	+0.042 [−0.00, +0.23]	0.949	+0.015 [−0.02, +0.11]
Llama-3.2-1B	BFCL	0.684	0.682	0.704	0.755	+0.077 [+0.01, +0.14]	+0.066 [+0.03, +0.10]	0.818	+0.014 [−0.05, +0.07]
Llama-3.2-1B	BFCL-live	0.685	0.653	0.680	0.682	+0.101 [+0.03, +0.18]	+0.075 [+0.03, +0.12]	0.795	−0.012 [−0.09, +0.06]
Llama-3.2-3B	Glaive	0.415	0.785	0.671	0.684	+0.227 [−0.02, +0.47]	+0.161 [+0.03, +0.32]	0.931	−0.020 [−0.11, +0.04]
Llama-3.2-3B	BFCL	0.681	0.702	0.627	0.633	+0.199 [+0.12, +0.27]	+0.142 [+0.10, +0.18]	0.758	+0.074 [+0.01, +0.14]
Gemma-3-1B	Glaive	0.599	0.519	0.869	0.872	+0.011 [−0.14, +0.13]	+0.021 [−0.05, +0.08]	0.853	+0.030 [−0.08, +0.17]
Gemma-3-1B	BFCL	0.916	0.856	0.882	0.900	+0.048 [+0.01, +0.08]	+0.039 [+0.02, +0.06]	0.927	+0.021 [−0.01, +0.05]
Qwen3-1.7B	BFCL	0.662	0.735	0.551	0.558	+0.187 [+0.05, +0.32]	+0.131 [+0.07, +0.19]	0.662	+0.084 [−0.02, +0.19]
MiniCPM5-2B	BFCL	0.641	0.779	0.557	0.585	+0.190 [+0.09, +0.29]	+0.131 [+0.08, +0.19]	0.677	+0.098 [−0.00, +0.21]
MiniCPM5-2B	BFCL-live	0.507	0.756	0.623	0.642	+0.102 [+0.01, +0.20]	+0.091 [+0.03, +0.15]	0.655	+0.090 [−0.01, +0.21]
Qwen3.5-0.8B	BFCL	0.663	0.838	0.660	0.678	+0.004 [−0.08, +0.08]	+0.031 [−0.01, +0.08]	0.730	−0.048 [−0.12, +0.02]

G. Every kind of error on every run

Purpose: the internal advantage of each kind of error against valid calls, on every run with at least twenty failures of that kind, and dropped calls scored within the parallel categories where both classes have twenty items. Wrong tool names reach twenty only on Llama-3.2-1B on BFCL-live, 68 of them, with an advantage of +0.216 [+0.06, +0.36].

Run	n_+	wrong values	n_+	missing arguments	n_+	dropped calls	dropped, parallel only
Llama-3.2-1B, Glaive	132	+0.357 [+0.00, +0.68]	86	+0.441 [+0.27, +0.63]	n.c.	n.c.	n.c.
Llama-3.2-1B, BFCL	127	+0.240 [+0.14, +0.34]	21	−0.055 [−0.18, +0.04]	n.c.	n.c.	n.c.
Llama-3.2-1B, BFCL-live	89	−0.010 [−0.20, +0.16]	46	+0.223 [+0.09, +0.36]	n.c.	n.c.	n.c.
Llama-3.2-3B, Glaive	85	+0.145 [+0.05, +0.32]	n.c.	n.c.	n.c.	n.c.	n.c.
Llama-3.2-3B, BFCL	88	+0.134 [+0.05, +0.22]	n.c.	n.c.	25	+0.158 [+0.05, +0.27]	+0.016 [−0.10, +0.13]
Gemma-3-1B, Glaive	109	+0.226 [+0.03, +0.36]	47	+0.260 [+0.09, +0.48]	n.c.	n.c.	n.c.
Gemma-3-1B, BFCL	33	+0.158 [−0.04, +0.33]	219	+0.188 [+0.14, +0.24]	99	+0.342 [+0.27, +0.42]	n.c.
Qwen3-1.7B, BFCL	33	−0.124 [−0.25, +0.00]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL	22	−0.162 [−0.30, −0.02]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL-live	84	−0.048 [−0.16, +0.04]	24	−0.085 [−0.28, +0.05]	n.c.	n.c.	n.c.
Qwen3.5-0.8B, BFCL	54	−0.121 [−0.22, −0.03]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL, JSON	21	−0.066 [−0.31, +0.12]	n.c.	n.c.	56	+0.201 [+0.15, +0.25]	+0.330 [+0.20, +0.45]
Qwen3.5-0.8B, BFCL, JSON	64	−0.136 [−0.23, −0.04]	n.c.	n.c.	143	+0.171 [+0.13, +0.21]	+0.198 [+0.09, +0.32]

H. Worked examples of each kind of error

Purpose: one stored generation per kind of error, the first in file order, so a reader can see what each label means. The first row is a value error by Gemma-3 that does not copy the schema. The second is a schema echo labelled missing arguments. A partial echo, which copies a type and a description into one argument without the schema’s outer fields, is not matched by

the echo pattern of Section 3 and can be labelled a value error.

Model	Mode	Ground truth	Generated call
Gemma-3-1B	wrong arg values	<code>[{"calculate_resonant_frequency": {"inductance": [0.05], "capacitance": [0.0001], "round_off": ["", 2]}]}</code>	<code>```json {"name": "calculate_resonant_frequency", "arguments": {"inductance": 50, "capacitance": 100, "round_off": 2 }}```</code>
Gemma-3-1B	missing args	<code>[{"algebra_quadratic_roots": {"a": [1], "b": [-3], "c": [2]}]}</code>	<code>{"name": "algebra_quadratic_roots", "arguments": {"type": "object", "properties": {"a": {"type": "integer", "description": "Coefficient of x^2."}, "b": {"type": ...</code>
Gemma-3-1B	missing calls	<code>[{"spotify_play": {"artist": ["Taylor Swift"], "duration": [20]}}, {"spotify_play": {"artist": ["Maroon 5"], "duration": ...</code>	<code>```json {"name": "spotify_play", "arguments": {"artist": "Taylor Swift", "duration": 20, "duration": 15 }}```</code>
Qwen3.5-0.8B, forced JSON	missing calls	<code>[{"spotify_play": {"artist": ["Taylor Swift"], "duration": [20]}}, {"spotify_play": {"artist": ["Maroon 5"], "duration": ...</code>	<code>{"name": "spotify_play", "arguments": {"artist": "Taylor Swift", "duration": 120}}</code>
Qwen3.5-0.8B	wrong arg values	<code>[{"solve_quadratic": {"a": [3], "b": [-11], "c": [-4], "root_type": ["all"]}]}</code>	<code><tool.call> <function=solve_quadratic> <parameter=a> 3 </parameter> <parameter=b> -11 </parameter> <parameter=c> -4 </parameter> <parameter=root_type> real </pa ...</code>
Qwen3.5-0.8B	valid	<code>[{"math_factorial": {"number": [5]}]}</code>	<code><tool.call> <function=math_factorial> <parameter=number> 5 </parameter> </function> </tool.call></code>

I. Confidence summaries

Purpose: the fold-chosen confidence summary of Section 6 in full.

On the re-extracted runs eleven length-invariant summaries of the token log-probabilities are stored per call: the mean, the minimum, the mean of the five least likely tokens, the perplexity, the fraction of tokens below one half, the mean, maximum and final predictive entropy, and the mean, minimum and mean-of-five over the call's own tokens with special tokens removed. For each fold the summary with the highest AUC on the items of the other folds is applied to the fold's items. On Llama-3.2-1B on BFCL the chosen summary reaches 0.666 against 0.651 for the mean, and the advantage becomes +0.165 [+0.09, +0.25]. On Gemma-3-1B on BFCL the chosen summary pools to 0.618, below the mean's 0.711, because the summary that leads on four folds trails on the fifth. On Qwen3.5-0.8B the chosen summary reaches 0.819 and the advantage −0.138 [−0.20, −0.07]. Two summaries that grow with the call's length, its token count and its total log-probability, are excluded as length features: on Gemma-3-1B on BFCL the token count alone reaches the length floor of that run, 0.842.

J. Head resolution controls

Purpose: the controls that attribute the per-head gain to head identity.

On the stored per-head features of 11 runs, under the same classifier and splits: the head-averaged graph's five metrics score 0.677, the same metrics computed per head and then averaged 0.709, all heads kept 0.790, the averaged features padded with Gaussian noise to the per-head width 0.568, and the per-head features with the head axis permuted independently per item 0.595. The gain of keeping heads apart over averaging the metrics is +0.081 and positive on 11 runs. Destroying head identity costs +0.195. Width without information changes the averaged read by −0.109. These controls train on every item and score the call-required population, the training population of the first draft, and both arms of each contrast share it.

K. Calls with a history

Purpose: the multi-turn protocol and its result.

The four multi-turn categories of BFCL-v4 (base, missing parameter, missing function, long context), one record per turn up to the fourth, with the ground-truth calls of the two preceding turns rendered as assistant messages, each followed by a fixed acknowledgement. In the corrupted condition one of the two preceding turns carries a call drawn from another conversation. Conversations with more than 22 tools are skipped. Folds group items that share either a tool or a conversation. Labels penalise arguments the schema does not define, because a weak model echoes tool schemas into argument-less calls on these prompts. On 538 turns of MiniCPM5-2B, 328 failures and 176 valid calls are scored. Linking tools and conversations

leaves four groups, so the folds and the group bootstrap rest on four units. The probe reads 0.900 and confidence 0.859, an advantage of +0.041 $[-0.01, +0.09]$, and +0.035 $[-0.02, +0.08]$ when the probe is trained on the scored population only.

L. Models and versions

Purpose: exact checkpoints and the layers read.

Table 3. The checkpoints, their release month, whether the chat template carries a reasoning mode (disabled at generation throughout) and the number of runs.

Checkpoint	Released	Thinking mode	Runs
Llama-3.2-1B	2024-09	no	3
Llama-3.2-3B	2024-09	no	2
Gemma-3-1B	2025-03	no	2
Qwen3-1.7B	2025-04	yes	2
MiniCPM5-2B	2026-09	yes	3
Qwen3.5-0.8B	2026-02	yes	1

Checkpoint	Released	Attention	Layers read
meta-llama/Llama-3.2-1B-Instruct	2024-09	dense GQA, 16 layers, 32 heads	all
meta-llama/Llama-3.2-3B-Instruct	2024-09	dense GQA, 28 layers, 24 heads	all
google/gemma-3-1b-it	2025-03	sliding window with 4 global layers, 4 heads	all
Qwen/Qwen3-1.7B	2025-04	dense GQA, 28 layers, 16 heads	all
Qwen/Qwen3.5-0.8B	2026-02	hybrid, softmax in 6 of 24 layers	softmax layers
openbmb/MiniCPM5-2B	2026-09	dense, 42 layers, 16 heads	all

Two further Qwen3.5 checkpoints in the stored data are left out: a 2B run with too few failures to score and a 4B run extracted before the stopping-token repair. Experiments were run on a single 16 GB consumer GPU, and the reader model and all comparators on CPU. Library versions, the generation seed and the commit are recorded in every extraction.